

Bliss Launcher

Feature documentation, version 8.3

A custom Android TV launcher for projectors, written in Kotlin with Jetpack Compose for TV. It is designed for a Magcubic projector running Android 14 (LuminaOS) with about 1 GB of RAM, and is operated with a remote control, a mouse, or both.

Item	Detail
Package	com.example.projectorlauncher
Version	8.3 (version code 74)
Target	Android 14 (targetSdk 34), minimum Android 8.3 (API 26)
Release file	dist/BlissLauncher-8.3.apk (signed, minified with R8)
Developer	Chetan Singh

1. Overview

Bliss Launcher replaces the stock home screen. It shows a clock and date, a row of quick buttons, a dock of chosen apps, and an All apps drawer. It can play a looping video wallpaper, start by itself at boot, and return to the front when the stock launcher appears. Every screen can be used with the D-pad, and with a mouse including hover, click, wheel and drag.

Main ideas

- **Light on memory.** Built for 1 GB of RAM: lazy decoders, caches on disk, no blur on low-RAM devices.
- **Never traps the user.** Back never closes the launcher; the highlight is restored after returning from an app.
- **Formal wording.** Polite, plain copy without contractions or dashes, following the owner's UI/UX document.
- **Fixed Midnight theme.** A single dark blue palette with consistent accents.

2. Home screen

Top bar

- Large clock and date at the top left, with an optional subtle CPU and RAM readout below the date. The readout reserves its space so nothing shifts when it appears.
- Five round quick buttons at the top right: Wi-Fi, Bluetooth, Screencast, Settings and Launcher settings. They carry no text labels and enlarge slightly when hovered or focused.
- The size of the quick buttons and the date can be changed together under Home screen (Top bar size).

Dock

- A translucent glass panel at the bottom holding only the apps you choose (at most six). All apps is always first, and a plus button opens a picker.
- The picker is a dialog with checkboxes over all installed apps.
- Icons show no names. The dock can be resized (Small, Normal, Large, Extra large).
- The dock is cached on disk, so it appears at once when returning from an app instead of reloading.

Behaviour

- Back on the home screen clears the highlight and never leaves the launcher.
- After opening something and returning, the previously focused button is highlighted again.
- If an app was opened from All apps, returning from it brings you back to All apps (automatic).
- The line "Developed by Chetan Singh" appears on the home screen only. A soft fade behind the text keeps it readable on bright wallpapers.

3. All apps

- Paged grid of 6 columns by 3 rows that fits the screen. Scrolling snaps one row at a time, so an icon is never cut by the border. The header shows the title and an app count badge.
- Apps are shown without tiles, like the dock, on a transparent background. With a video wallpaper, the drawer can use the static wallpaper in Midnight tones instead (Static wallpaper in app drawer).
- Press and hold (or right-click) an app to open a menu with **App info** (opens the system page for Force stop, Uninstall and so on) and **Hide from All apps**.
- Hidden apps are managed under Launcher settings, Hidden apps, where one action restores all of them.

- The app list is cached on disk so it is shown immediately and refreshed quietly. A package-change receiver updates it when apps are installed or removed.

4. Mouse and remote support

Input	Behaviour
D-pad / remote	Full navigation of every screen. Enter selects. Back goes up one level and never closes the launcher.
Mouse hover	Moves the highlight to the item under the pointer. Only real mouse movement counts, only on fully visible rows.
Mouse click	Opens the item. Press and hold behaves like a long press.
Wheel and drag	Scroll lists. Moving the pointer to the screen edge never scrolls a list.
Magnification	The hovered or focused icon grows slightly on all sides. Neighbours do not move.

5. Video wallpaper

A muted, looping video can play behind the home screen. It is configured under Launcher settings, Video wallpaper.

Library and selection

- **Folder.** Choose the folder with the file manager, or use the app folder (no permission needed). If the device has no folder picker, selecting any video derives its folder.
- **Thumbnails and list views.** Small thumbnails, and three views: Usable only, Selected only and All videos. Select all usable and Clear selection act on the whole list, so long folders need no manual checking.
- **Rules.** Landscape only, no stretching, at most 30 seconds, and the picture must fit the screen without borders (aspect ratio within 2 percent). Videos too large for the hardware decoder are marked TOO LARGE. An optional filter allows only 120 fps clips.
- **Formats.** MP4 and WebM are supported. Sound is always off.
- **Several videos** play in turn with a dissolve between them. One video loops with a dissolve onto itself.

Playback options

Option	Effect
Video wallpaper	Master switch.
Hide icons after	Idle time before the icons fade away so the wallpaper is fully visible (never is possible).
Pause while icons are visible	The video holds still while icons show and plays when they fade out.
Smooth pause and resume	Eases to a stop and back up (see below). When off, the video stops and starts at once.
Playback speed	0.25x to 2x. 0.5x, 1x and 2x give the most even pacing on a 60 Hz display.
Static wallpaper in app drawer	All apps shows the static wallpaper instead of the video.

How smooth pause and resume works

The video never creates new frames; changing the speed only changes how long each source frame is held. Because a 24 or 30 fps clip looks stepped at very low speeds, the pause stops at a cutoff speed and then performs a real pause (the picture stays on screen, no seek, no decoder restart).

- **Cutoff speed** = 1000 / (frames per second x 100), limited to 0.25x to 0.50x. About 0.33x at 30 fps, 0.42x at 24 fps, and 0.4x when the frame rate is unknown.
- **Pause:** a 2.2 second slowdown from the current speed to 1.15 times the cutoff, braking late (progress follows t squared), then the player is paused.
- **Resume:** playback starts at the cutoff and rises to full speed in 0.6 seconds (progress follows 1 minus (1 minus t) squared).
- **Timing** uses the monotonic clock, so a late callback samples the curve where it is now. Only one ramp runs at a time and stale completions are ignored. A reversal starts from the latest speed.
- **Dissolves:** the current speed is carried to the incoming clip; a pause requested during a dissolve is applied when the dissolve ends.
- **Leaving the launcher** (another app opens, screen off) pauses the video at once with no ramp.

Tuning values are constants in VideoWallpaper.kt: PAUSE_MS, RESUME_MS, PAUSE_STOP_FACTOR, MAX_FRAME_GAP_MS and the cutoff limits. Debug builds log ramp timing under the tag VideoRamp.

Memory design

- Only one decoder runs normally. A second is created about 3 seconds before a dissolve and released right after it.
- If a second decoder cannot start, the launcher switches to a single-decoder mode that cuts between clips instead of dissolving.
- Small load-control buffers, audio disabled, everything released when the launcher stops.

The helper script tools/smooth-slowmo.sh can bake a 60 fps interpolated copy of a clip with ffmpeg for smoother slow speeds.

6. Launcher settings

Opened from the rightmost quick button. The sidebar is only as wide as its longest name, and sections follow a logical order. Switches replace ON and OFF labels, and each row has a short subtitle.

Section	Contents
Home screen	Dock size, top bar size (quick buttons and date together), CPU and RAM readout.
Wallpaper	Choose a still image (.jpg, .png, .webp) from the wallpaper folder; refresh list.
Video wallpaper	All video options described in section 5.
Button targets	Destination of Settings, Wi-Fi and Bluetooth: Automatic, or any activity. The package list leads to the activities of a package, or a class name can be entered. Save and test opens the target for verification. Defaults point to the TV settings (com.android.tv.settings) with fallbacks; a custom activity such as com.htc.luminaos.activity.MainSettingActivity can be entered.
Screencast app	The app opened by the Screencast button. Detected automatically (any installed app, including third party) or chosen manually.
Shortcuts	Direct entries for Developer Options, About Device and Install Unknown Apps.
Hidden apps	List of hidden apps; restore one or all.
Automation	Start the launcher at boot, keep it on top, with a pause option for temporarily using the stock launcher.
Stock launcher	The package of the system launcher to monitor (detected, or entered manually).
Permissions	Display over other apps, Usage access and notifications, each with a shortcut to the system page.
Diagnostics	Lists the installed settings packages, useful when a button does not open.

7. Automation

- **Start at boot.** A boot receiver reads settings from device-protected storage, so it works before the first unlock, and starts the launcher after a delay.
- **Keep my launcher on top.** A foreground service watches the foreground app through usage statistics. When the stock launcher appears it brings Bliss Launcher back. A quick pause switch allows temporary access to the stock launcher.
- **Requirements.** Display over other apps and Usage access must be granted (both have shortcuts in Permissions).
- **Safety.** Each action has a try and catch with an on-screen failure message such as "Unable to open ...", and falls back from the standard action to specific components and then a generic page.

8. Performance on 1 GB of RAM

- Low-RAM mode: 565 wallpaper bitmaps and no blur on glass panels.

- Icon LRU cache, plus disk caches for the app list and the dock, avoid a visible reload after returning from an app.
- Work stops when the launcher is not resumed.
- Release builds use R8 minification and resource shrinking; a release-like profile build type exists for measurements.
- Measured on a 1 GB emulator: about 1.4 decoders on average instead of 2, no crashes in monkey stress testing.

9. Look and branding

- Midnight palette, glass dock and top buttons that are translucent without reflections.
- Adaptive launcher icon with a monochrome variant, TV banner, and an animated splash mark (Android 12 and later).
- Master artwork in the branding folder; a 512 px store icon and lockup in dist.

10. Installing and building

Install on the projector

- Copy dist/BlissLauncher-8.3.apk to the projector and install it with a file manager (Install unknown apps must be allowed for that file manager). Installing over version 7.5 keeps settings, because both use the same signature.
- Press Home and choose Bliss Launcher as the home app. Under Launcher settings, Permissions, grant Display over other apps and Usage access for the keep-on-top feature.
- Install a new build only with the same signing key; otherwise Android refuses the update.

Build from source

- ./gradlew :app:assembleRelease for the signed release (keystore.properties and the keystore folder are required).
- ./gradlew :app:assembleProfile for the release-like debuggable build used for measurement and logging.
- Keep keystore.properties and the keystore file safe and private; losing them means a different signature.

11. Known limitations

- USB debugging on the target projector is locked, so on-device behaviour (boot start, folder picking, decoder behaviour, mouse hover) was verified on an Android 14 TV emulator with 1 GB of RAM only and should be confirmed on the projector.
- Very slow speeds on 24 and 30 fps clips look stepped; a 60 fps baked copy helps.
- Videos larger than the hardware decoder limit (for example 4K) are blocked.
- Android 14 may offer only partial video access; choose the app folder or allow all videos if the list is empty.